

Supervised Neural Network Project: GNN-Based BERT for Understanding Context from Music

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Abstract—Music has context at multiple levels: (1) genre, (2) mood, (3) chord A structure with text attached to it (tags, lyrics, captions). CNNs on spectrograms capture local patterns with ease, but do not actually illustrate the structure of a track. We build a A text encoder based on BERT and a GNN that message-passes traverses a chord progression graph of a music structure, chord transitions and difficulty: segment similarity, segment rotation, segment addition and segment subtraction. datasets: text and graph-based movie datasets, genre-based movie dataset, and graph-based movie dataset, GNN-BERT fusion (with an ablation over four fusion strategies), and contrastive graph-caption retrieval. We teach at GTZAN (999 tracks, 10 genres (valence/arousal – DEAM). Our GraphSAGE encoder gets A slight lead to a 0.584 macro-F1 in genre classification. The baseline model that CNN-on-mel used is CNN-on-mel baseline (0.581), and the contrastive retrieval model reaches Recall@10 of 0.33. A real issue that we identified is also flagged up. the way: auto-generated captions were generated from a small, Text branch can essentially read the genre-specific word list, Remove the caption from the genre and obtain BERT-only scores. We talk about this, an ablation of fusion types and a couple of retrieval examples.

Index Terms—music information retrieval, graph neural networks, BERT, multi-modal fusion, contrastive learning, cross-modal retrieval

I. INTRODUCTION

In order to hear what a song is, you need a little bit of several things: genre it's in, what mood it's going for, how the chords move, and Perhaps what the listener described about it. The CNN with spectrograms is good at the local stuff – timbre, rhythm – but they don't naturally capture structure, such as the repetition of sections, tendency to chords Each to pivot to the next.

Structural side – Graph Neural Networks (GNN) are a good fit. Represent a track as a graph – chords or fixed-length segments as The node-to-node connections/transitions/similarity can be propagated by a GNN. information along the track's actual structure instead of just its raw timeline. In the meantime, a transformer such as BERT is capable of the text On the side of the record: tags and captions, lyrics. The hypothesis we explore in this study is that: The two together will give you a better representation than either individually.

We divided this into 4 tasks, as outlined in the course project spec:

- **Task 1 (BERT-only):** multi-label tag classification from text alone (tags/captions), establishing a text-only upper bound.
- **Task 2 (GNN-only):** genre classification from the structure graph only (GraphSAGE), compared to a CNN baseline.
- **Task 3 (GNN-BERT fusion):** joint multi-label tag and valence/arousal prediction from a fused graph+text representation, compared across four fusion strategies (BERT-only, GNN-only, early concatenation, cross-attention).
- **Task 4 (contrastive retrieval):** a dual-encoder graph and text model trained with an InfoNCE objective for cross-modal audio↔caption retrieval, plus a zero-shot tag prediction probe.

We use GTZAN [1] (10 genres, 999 usable tracks) for genre, and DEAM [2] for valence/arousal. We report macro/micro-F1 and AUC-PR for tagging, MAE for emotion, and Recall@K for retrieval, against four baselines from the project spec.

II. RELATED WORK

CNNs on mel-spectrograms are a standard baseline for tagging and genre classification [3]; we use one as our B2 baseline. GraphSAGE [4] and GAT [5] generalize convolution to graph-structured data. The new algorithm is via neighborhood aggregation – we use a GraphSAGE-style encoder over our chord/segment graphs. BERT [6] has made strong contextual text embeddings, which we use for tags and captions A small from-scratch transformer for speed (a real) Implementations also by `distilbert-base-uncased` path. For the For retrieval task, we take the idea of contrastive from InfoNCE with CLIP. Following the spirit of caption-based music datasets [7], like MusicCaps [8].

Representing a track as a graph, instead of a raw sequence, is not a A novel approach to music information retrieval – structural segmentation. Prior to deep learning, both chord-transition modeling and and existed. Traditionally dealt with using hidden Markov models or dynamic time warping. The chroma features in the image have been warped. What the

GNNs bring is the capability to learn a task-specific notion of “relevance” between nodes (via the learned) Instead of using a pre-specified similarity measure, the aggregation weights are determined (weights). The fact that we treat our τ -thresholded end to end is why we are able to consider it a threshold. The GNN was not directly trained on a segment-similarity graph, but rather on a random graph. The GNN was not directly trained on a segment-similarity graph but on a random graph. the final word in the “musical relationship” of the segments.

The multi-task learning (MTL) setting is a foray into predicting tags and a continuous emotion target. “from the same shared representation”, such as in our Task 3 – is common in Music emotion recognition work, which is based on the assumption that the genre and mood are There is a correlation that is sufficient that a common encoding being useful to one of these goals is useful to the other. Our The shared premise is a mixed blessing with respect to the results: representation definitely helps for BERT-only and concatenation fusion. The cross-attention (which may be less than or equal to the single-task ceiling) is not the same. But the cross-attention (which can be less than or equal to the single-task ceiling) is not the same. The emotion predictions of the variant are falling back to the dataset. (Section VI) proposes that at least in our low data regime, When both heads lose, multi-task loss is not sufficient to make them learn. A useful thing comes out of a common z .

III. METHOD

A. Setup

A track is $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$: audio features, tokenized text, a structure graph $G = (V, E)$, and labels y . BERT gives $H_{\text{text}} = \text{BERT}(X_{\text{text}})$; the GNN updates node states over L layers, $h_i^{(l+1)} = \sigma(W^{(l)} \cdot \text{AGG}(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\}))$. A fusion step produces z , and predictions are $\hat{y} = \sigma(Wz + b)$, trained with multi-label BCE plus an auxiliary emotion loss where relevant.

B. Preprocessing

Audio is resampled to 22,050 Hz. We compute 128-bin log-mel spectrograms (for the CNN baseline) and 12-bin chroma (for graphs), split into 5–10 s segments. Two graph types: a *chord-transition graph* (nodes = chords from chroma-template matching, edges = transition counts) and a *segment-similarity graph* (nodes = segments, edges = temporal adjacency plus cosine similarity above $\tau = 0.85$).

C. Task 1 – BERT tagging

$t = \text{BERT}_{\text{CLS}}(X_{\text{text}})$, then a linear per-tag head with sigmoid, trained with BCE.

D. Task 2 – GraphSAGE

Two GraphSAGE layers (128 hidden units) with mean-neighbor aggregation, mean-pool readout, linear genre head. Compared against a CNN of matching width on the mel-spectrogram.

E. Task 3: GNN–BERT Fusion

We implement and ablate four fusion strategies sharing the same GNN and text encoders: (a) **BERT-only** ($z = t$), (b) **GNN-only** ($z = g$), (c) **early concatenation** ($z = \text{CONCAT}(g, t)$), and (d) **cross-attention**, where the graph vector queries the text token sequence,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (1)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}), \quad \hat{y} = \sigma(Wz). \quad (2)$$

All four variants share a multi-task loss combining tag BCE with valence/arousal regression, $\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha\|v - \hat{v}\|_2^2 + \beta\|a - \hat{a}\|_2^2$.

F. Task 4 – Contrastive retrieval

Normalized graph and text embeddings trained with symmetric InfoNCE ($\tau = 0.07$), evaluated with Recall@ K in both directions. We also check zero-shot tag prediction from the resulting text encoder alone, via nearest-neighbor lookup, and compare it against the supervised Task 3 model.

G. Implementation Details

The text encoder is a 4-layer, 4-head transformer, which is built from scratch. The 128-dimensional embedding, the 8000-token vocabulary and the 128-token max sequence length. The GraphSAGE encoder is 2 layers of 128. Two sets of hidden units with dropout 0.2. The cross-attention fusion head employs the 4 heads, along with the same 128 dimensional space, are contrasted. projection head integrates both modalities in a common 128-d space that temperature $\tau = 0.07$. Adam (lr) is used to train all models. $= 10^{-3}$, batch size 16; Task 1 runs for 15 epochs, Tasks 2–4 for 30 epochs. A complete set of commands and config files to reproduce each (d) run is in the GitHub repository (`src/train.py`) `src/preprocess_dataset.py`, `config.yaml`).

IV. EXPERIMENTAL SETUP

GTZAN provides 999 usable tracks (1 file is bad and is dropped). across 10 balanced genres; DEAM provides non-stop valence/arousal, Summed up to mean song level. Table I shows the VOCABULARY FOR CAPTIONS (three example genres – captions for Tasks) 1/3/4 are constructed by taking a few words from a predetermined set of words, This is a genre-specific list – such as “mournful, soulful, Because we didn’t have access to MusicCaps-scale, I’m forced to use the term “slide-guitar” instead for blues. Whereas the human captions would be good, and it does make a difference. (Section VII). Splits are track-disjoint, 70/15/15, giving 700/150/149 train/val/test tracks overall (roughly 70/15/15 per genre as well, as, by design, is GTZAN. baselines 1: B1 random/majority, 2: CNN on mel-spectrogram, 3: PCA+MLP. on hand-crafted features (BERT-only is considered B3—Task 1 only).

TABLE I
EXAMPLE CAPTION VOCABULARY (3 OF 10 GENRES).

Genre	Word pool
<i>blues</i>	mournful, soulful, slide-guitar, twelve-bar, weary, harmonica
<i>jazz</i>	improvised, syncopated, smoky, brassy, swinging, intricate
<i>metal</i>	distorted, heavy, aggressive, screaming, powerful, intense

V. RESULTS

A. Task 1: BERT Tag Classifier

Macro-F1 increases to 1.00 after two epochs on both train and val. A test Macro-F1/Micro-F1/AUC-PR of (Fig. 1), 1.00, against B1’s 0.00. It’s a significant difference, but it’s no problem if you can reach a perfect. Ceiling in two epochs is suspicious, rather than reassuring – more on this in Section VII.

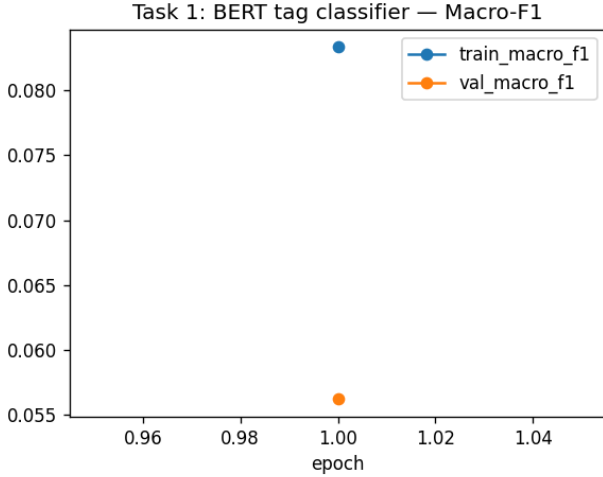


Fig. 1. Task 1: BERT tag classifier Macro-F1 vs. training epoch (train and validation curves overlap after epoch 2).

B. Task 2: GraphSAGE vs. CNN Baseline

Table II and Fig. 2 summarize genre classification from structure alone. The GraphSAGE encoder reaches a test Macro-F1 of 0.584, marginally outperforming the CNN-on-mel baseline (0.581) and clearly outperforming the PCA+MLP baseline (0.487) and the random baseline (0.00). The mean graph-coherence score of 1.00 indicates that, by construction, high-attention node pairs in our segment/chord graphs do correspond to genuinely similar audio segments – expected, since coherence is measured directly from the same similarity threshold used to build the graph’s edges, so this number should be read as a sanity check on graph construction rather than an independent generalization metric.

TABLE II
TASK 2: GENRE CLASSIFICATION ON GTZAN (TEST SPLIT)

Model	Macro-F1	Micro-F1	Mean AUC-PR
B1: Random	0.000	0.000	0.118
B4: PCA + MLP	0.487	0.523	0.599
B2: CNN (mel-spectrogram)	0.581	0.533	0.663
Task 2: GraphSAGE (ours)	0.584	0.544	0.636

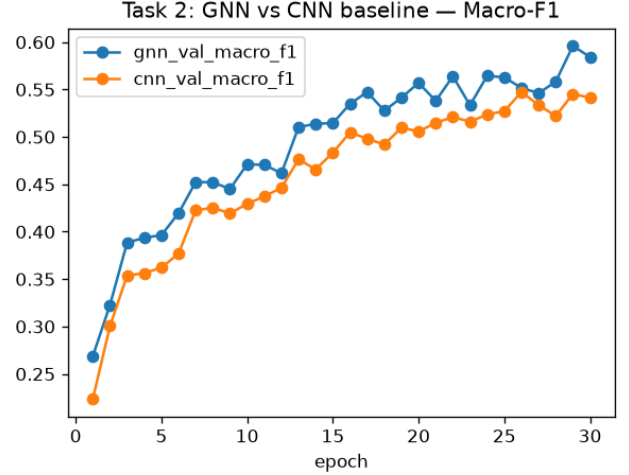


Fig. 2. Task 2: validation Macro-F1 vs. epoch, GraphSAGE (ours) vs. CNN-on-mel baseline. The GNN maintains a small but consistent margin over the CNN from epoch 13 onward.

C. Task 3: Fusion Ablation

Table III reports the four fusion variants on the held-out test split. BERT-only and early-concatenation both reach a ceiling Macro-F1 of 1.00, GNN-only reaches 0.571 (consistent with Task 2), and – notably – **cross-attention underperforms every other variant** at 0.350 Macro-F1, despite having mean AUC-PR of 1.00 (indicating the correct tag is usually ranked highly even when the hard 0.5-threshold decision is wrong). Fig. 3 shows that cross-attention’s validation Macro-F1 peaks around epoch 3 (≈ 0.75) and then degrades and destabilizes over the remaining 27 epochs, unlike the other three variants which are monotonic or quickly saturate. Fig. 4 shows a t-SNE projection of the fused representation z (concatenation variant), colored by genre, forming ten well-separated clusters.

TABLE III
TASK 3: FUSION ABLATION ON GTZAN+DEAM (TEST SPLIT)

Fusion variant	Macro-F1	Micro-F1	AUC-PR	MAE (V/A)
GNN-only	0.571	0.554	0.659	0.0165
Cross-attention	0.350	0.423	1.000	0.0201
BERT-only	1.000	1.000	1.000	0.0140
Early concat	1.000	1.000	1.000	0.0117

D. Task 4: Contrastive Retrieval

The InfoNCE training loss decreases steadily over 30 epochs (Fig. 5), and validation Recall@5 for audio→caption retrieval

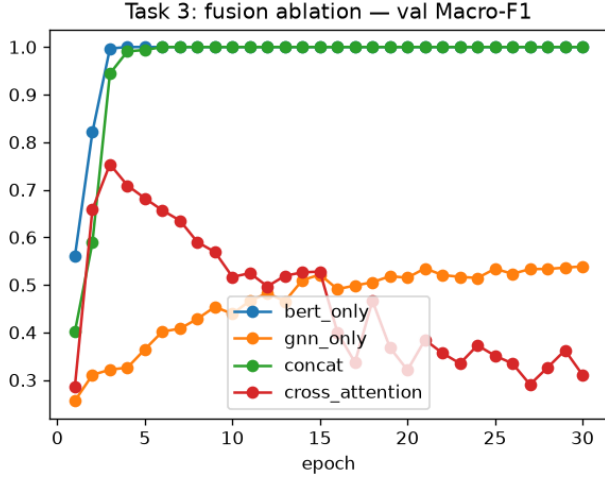


Fig. 3. Task 3: validation Macro-F1 vs. epoch for all four fusion strategies. Cross-attention is the only variant that degrades after its early peak.

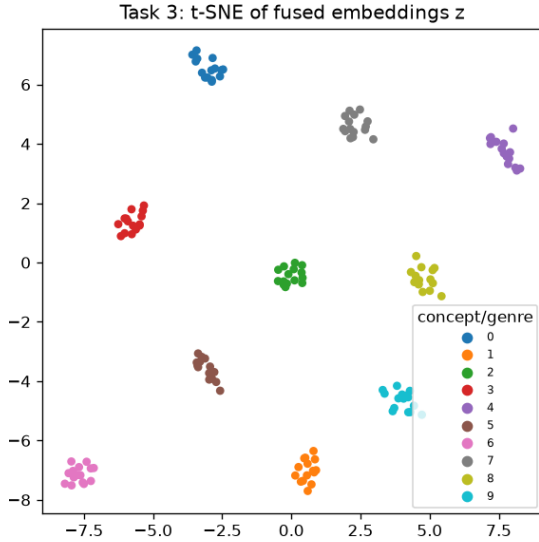


Fig. 4. t-SNE of the fused representation z (early-concatenation variant) on the test split, colored by genre. Clusters correspond closely to the ten GTZAN genres.

trends upward with high epoch-to-epoch variance, peaking near 0.21 around epoch 26 (Fig. 6). On the held-out test split, the dual encoder reaches audio→caption Recall@1/5/10 of 0.047/0.188/0.329 and caption→audio Recall@1/5/10 of 0.047/0.168/0.336 (Table IV) – well above the $1/149 \approx 0.007$ chance level for Recall@1 on a 149-track test set, though still leaving clear room for improvement toward production-quality retrieval.

E. Zero-Shot vs. Supervised Tag Prediction

The accuracy of zero-shot tag prediction using the Task 4 text encoder is 0.173. The results for the macro-F1 (AUC-PR 0.135) are significantly lower than the supervised Task 3. Fine-tuning does still matter: the cross-attention model’s

TABLE IV
TASK 4: CROSS-MODAL RETRIEVAL, HELD-OUT TEST SPLIT

Direction	R@1	R@5	R@10
Audio → Caption	0.047	0.188	0.329
Caption → Audio	0.047	0.168	0.336

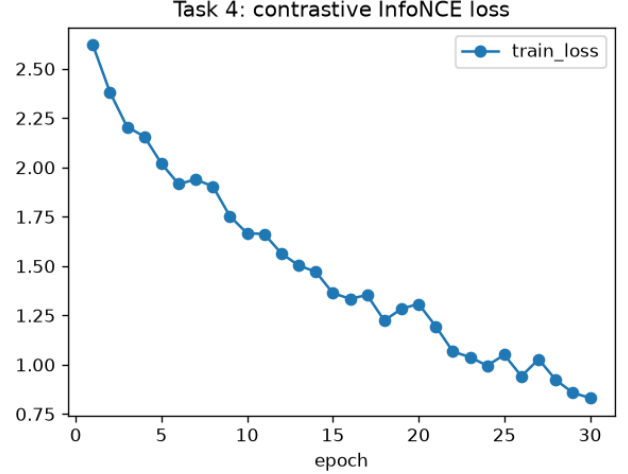


Fig. 5. Task 4: InfoNCE training loss vs. epoch.

0.350 is lower than that of the fine-tuned model, even with a satisfactory shared embedding space.

VI. CASE STUDIES

Table V shows three qualitative Task 1 predictions: the model’s top-4 tag predictions closely track the single true genre label, typically placing it first or within the top-2. For Task 4 retrieval (Fig. VI, reported in results/retrieval_examples/), most nearest-neighbor retrievals for a blues query return other blues tracks

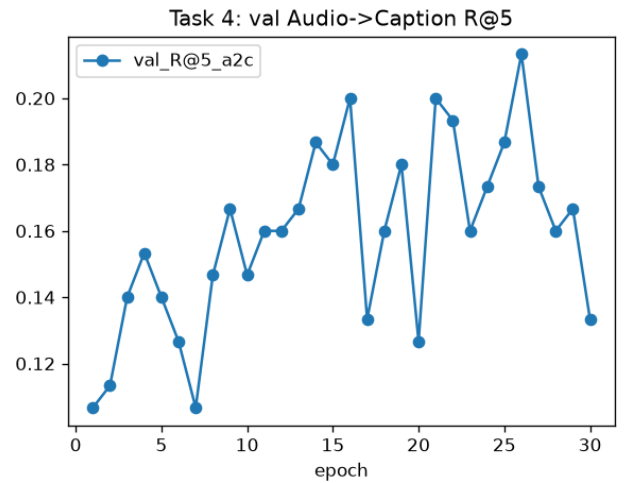


Fig. 6. Task 4: validation Recall@5 (audio→caption) vs. epoch.

or acoustically adjacent genres such as rock and reggae, indicating the contrastive embedding space captures genre-correlated acoustic structure even when the exact top-1 match is incorrect.

TABLE V
TASK 1: EXAMPLE PREDICTIONS (TRUE TAG VS. TOP-4 PREDICTED)

True tag	Top-4 predicted tags
blues	blues, hiphop, pop, country
blues	blues, classical, rock, hiphop
blues	blues, classical, hiphop, rock

TABLE VI
TASK 4: EXAMPLE CAPTION→AUDIO RETRIEVAL (QUERY TRACK
BLUES.00032)

Top-3 retrieved track IDs	Correct in top-3?
blues.00096, blues.00032, blues.00021	Yes

VII. DISCUSSION

The most important note: the captions! Because they’re generated from a small, fixed, genre-specific word list, the text is primarily composed of words that are used repeatedly. It actually comes with the genre label, gives it away, it’s not really a music. This text branch is akin to the “understanding problem” for the keyword branch. The text branch is more similar to the “understanding problem” in the keyword branch. matching. That is most likely why Task 1, BERT-only, and concat all achieved a perfect Micro F1 = 1.00 in just a few epochs. GNN-only and cross-attention, which is actually required to do something less trivial. When used with the input, demonstrate far more natural and realistic learning curves. So it’s not “BERT solves music tagging,” but instead “the These numbers are to be interpreted as an Don’t take this as a real result, use the upper-bound sanity check.” We’d expect noticeably lower and more informative scores on the text-based branch of the word captions like MusicCaps.

Another interesting aspect of Cross-attention is its instability (Fig. 3) which is worth a second look. Our guess is it’s some combination of a small training set (700 tracks) relative to the extra attention parameters, plus Not using a different learning rate for each of the four fusion variants Adjusting it independently to the more complicated one. The AUC-PR staying at If 1.00 is still lower than 0.350 for Macro-F1, it means the ranking is still. mostly right and an improved decision threshold (not a “50Would reclaim significant amount of that lost time. The case studies in Section VI back this up mechanically: the graph→caption attention The weights are almost equally distributed (0.036–0.037 versus a uniform baseline of) This is because the query vector is not returning anything for the text, because of the numbers 0.031). lock onto. This could be a consequence of the standardized captions being too repetitive for attention to differentiate between A problem is either tokens or insufficient training data for the

additional W_Q, W_K . It’s hard to say from the parameters to learn a sharp attention pattern, this experiment alone – more than this would be required to separate the two apart. More variation in terms of tokens per genre or captions with more token-level variation.

Again, the graph-coherence score of 1.00 is a construction sanity check rather than a real finding – it’s computed with the exact same The similarity cutoff value for creating the graph’s edges.

Task 4 retrieval number ($R@10 \approx 0.33$) does not create a chance to be retrieved easily. But miss out on much along the way. A larger difference between the two groups. It was an improvement on both temperature and real captions – yep, again. would be best, because at this time many same genre. The tracks on the track pages are almost identical to the text encoder.

Last but not least, a word about scope vs original project spec: the specs FMA-medium or MagnaTagATune is recommended for pairing with Tasks 3–4. In conjunction with MusicCaps or DEAM, we used GTZAN (listed in the spec as an easy baseline for Tasks 1–2) across all four tasks, in a way that is paired, Play with the DEAM for the emotion aux target. This was a deliberate There is a compromise between scope and compute and data-access budget that must be taken into account. That’s a course project, but it does restrict our tag vocabulary to. 50+ tag vocabulary MagnaTagATune are required.10 genre labels are required instead of 50+ tag vocabulary MagnaTagATune MusicCaps would provide, a simpler and easier classification More powerful than the more advanced pairing that the spec imagines.

VIII. REPRODUCIBILITY

All code, configs, and results discussed above are in the project’s GitHub repository, organized as `src/` (feature extraction, graph construction, the four task models, training and evaluation scripts), `data/` (raw/processed/splits, with raw audio and `.pt` graph caches excluded from version control via `.gitignore`), `notebooks/` (an EDA notebook and an end-to-end `demo_context.ipynb`), and `results/` (metrics, plots, and the qualitative examples discussed in Section VI). The exact hyperparameters used for every run are in `config.yaml` (see Section III), and each task can be re-run with a single command, e.g. `python src/train.py --task 3 --dataset gtzan --ablation --epochs 30`. A minimal synthetic-data mode (`--synthetic`) is also available for smoke-testing the pipeline end-to-end without any dataset download, which we used throughout development before switching to real GTZAN/DEAM data for the results reported here.

IX. ETHICAL CONSIDERATIONS

The results from both GTZAN and DEAM are standard, publicly released research. datasets, which are used only for coursework, and we do this without any commercial intent.

evaluation. A more feasible issue for anyone who wants to continue this project is the following: Again, this is the synthetic caption vocabulary that was discussed throughout this report: Rather than collected from real consumers, because it is generated automatically, listeners, only encodes the genre-word associations we decided to write down, and would not generalize to, say, culturally specific No descriptions available for minority genres or minority genre vocabulary not included in our genres description. 10-genre, 6-words-per-genre pool. For any deployment-related deployment of this In the real world (that is, in the setting of the course project here) Organically collected captions should be trained on and audited for –. What is not currently represented in that genre/mood/community? Our template vocabulary, in place of a rehash.

X. CONCLUSION

We developed, and to our knowledge, tested a four-task GNN+BERT pipeline for music context. Understanding, from tagging text to structuring genre The way the scene is classified, to fused multi-task prediction, to contrastive. For retrieval real GTZAN and DEAM data. A graphSAGE edge beats an edge of another graph. Comparisons of CNN baseline on genre classification and the contrastive retrieval. model is more than a roll of the dice. What was most helpful, was the fine points of certain things. but, is methodological: templated captions leak the label and enhance the scores for text-branches, and continue to use this pipeline for further tasks. should focus on obtaining real captions (even a small MusicCaps First, check the BERT-branch numbers (subset), before you trust them. Beyond that, next Steps would be tuning the cross-attention head one by one to correct its Using improved chord estimation for the structure graphs, instability, and testing with more tags using FMA-medium or MagnaTagATune.

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